Question ID: 1087f6c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

$$24.5x + 24.75y = 641$$

Isabel ordered topsoil and crushed stone, which cost a total of $\$641$, for her garden. The given equation represents the relationship between the number of cubic yards of topsoil, x, and the number of tons of crushed stone, y, Isabel ordered. How much more, in dollars, did a ton of crushed stone cost Isabel than a cubic yard of topsoil?

Correct Answer: 0.25, 1/4

Rationale

The correct answer is $.25$. It's given that the topsoil and crushed stone Isabel ordered for her garden cost a total of $\$641$. It's also given that the equation $24.5x + 24.75y = 641$ represents the relationship between the number of cubic yards of topsoil, x, and the number of tons of crushed stone, y, that Isabel ordered. Since x represents the number of cubic yards of topsoil ordered, $24.5x$ represents the total cost, in dollars, of the topsoil, and the cost per cubic yard of topsoil is $\$24.50$. Similarly, since y represents the number of tons of crushed stone ordered, $24.75y$ represents the total cost, in dollars, of crushed stone ordered, and the cost per ton of crushed stone is $\$24.75$. Therefore, a ton of crushed stone cost Isabel $24.75 - 24.50$, or 0.25, more dollars than a cubic yard of topsoil. Note that .25 and 1/4 are examples of ways to enter a correct answer.